

CS 342: Computer Vision and Pattern Recognition

Instructor: Br. Tamal Mj

Shanti Mantra

Lyrics in Sanskrit

Meaning in English

ॐ सह नाववतु।

Om, May we all be protected

सह नौ भुनक्तु।

May we all be nourished

सह वीर्यं करवावहै।

May we work together with great energy

तेजस्यि नावधीतमस्तु

May our intellect be sharpened (may our study be effective)

मा विद्विषावहै।

Let there be no Animosity amongst us

ॐ शान्तिः शान्तिः शान्तिः ॥

Om, peace (in me), peace (in nature), peace (in divine forces)

Jean Baptiste Joseph Fourier (1768-1830)

had a crazy idea (1807):

Any univariate function can be rewritten as a weighted sum of sines and cosines of different frequencies.

- Don't believe it?
 - Neither did Lagrange, Laplace, Poisson and other big wigs
 - Not translated into English until 1878!

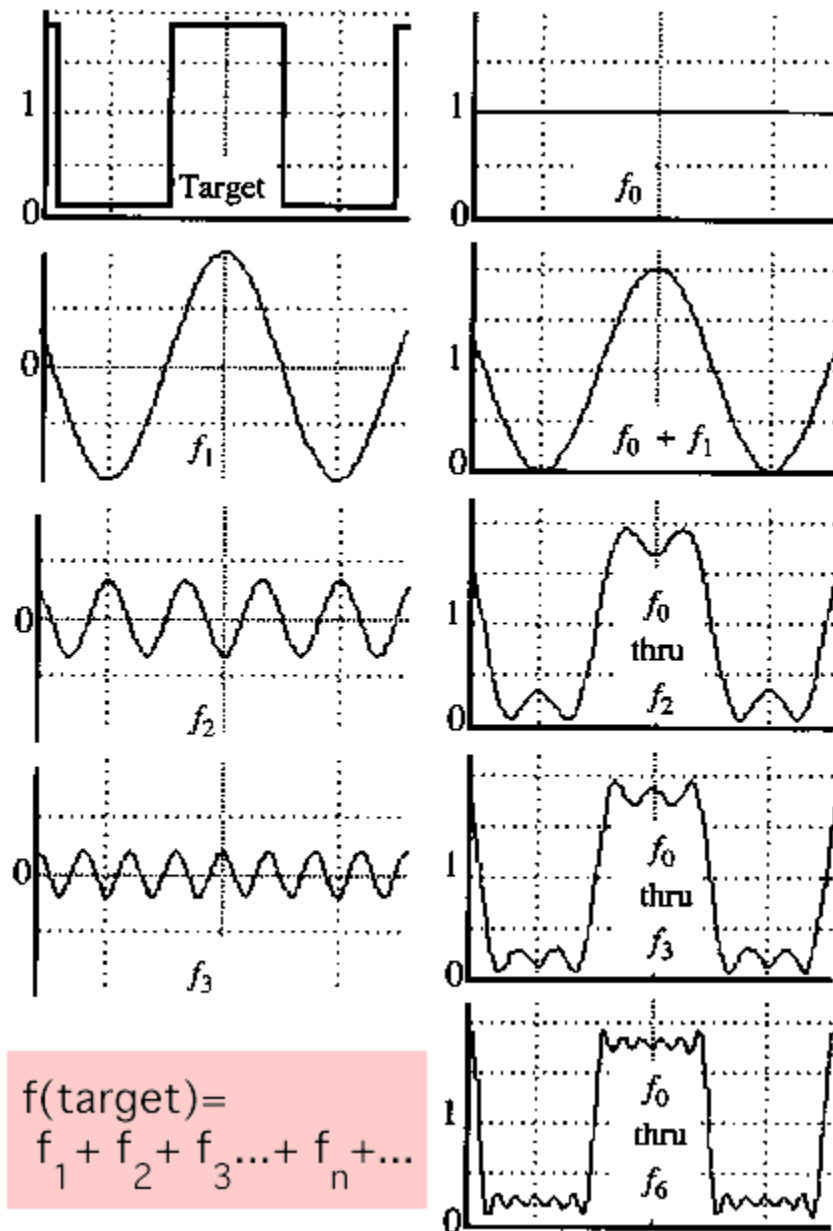


A sum of sines

Our building block:

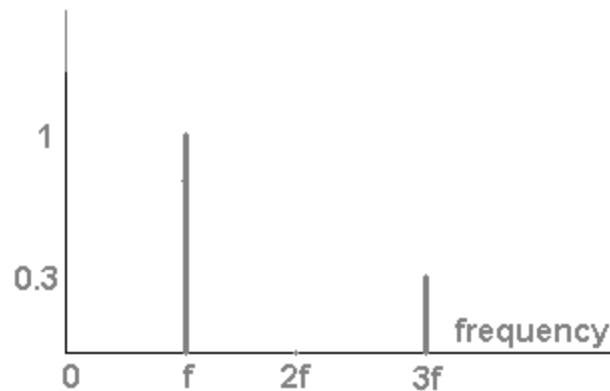
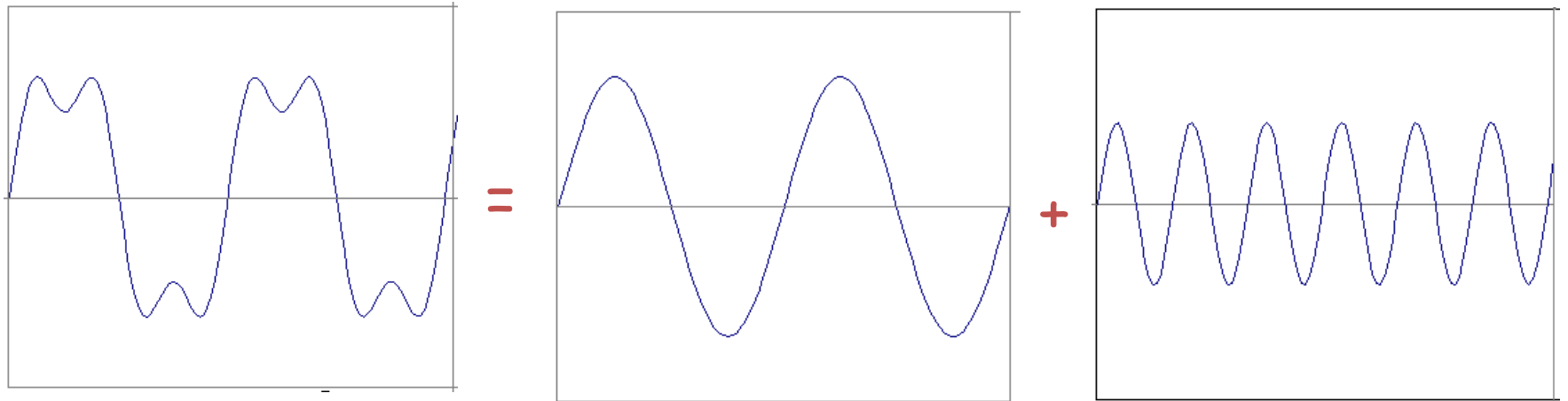
$$A \sin(\omega x + \phi)$$

Add enough of them to get any signal $g(x)$ you want!

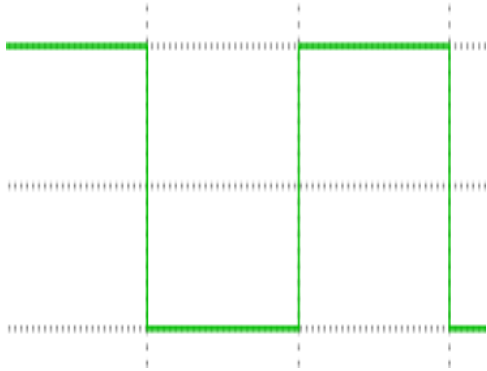


Frequency Spectra

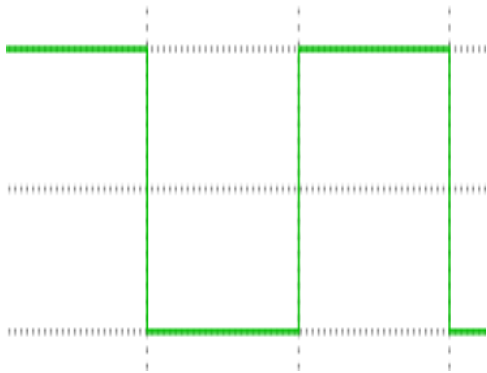
- example : $g(t) = \sin(2\pi f t) + (1/3)\sin(2\pi(3f) t)$



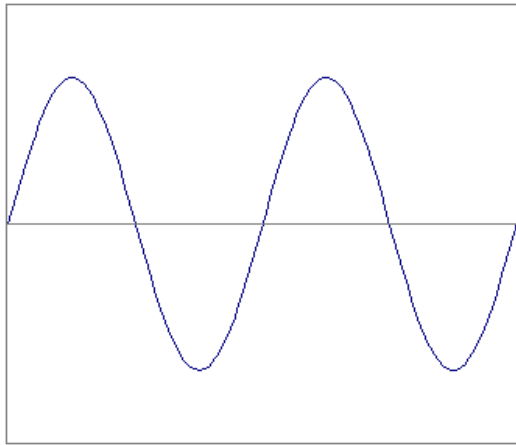
Frequency Spectra



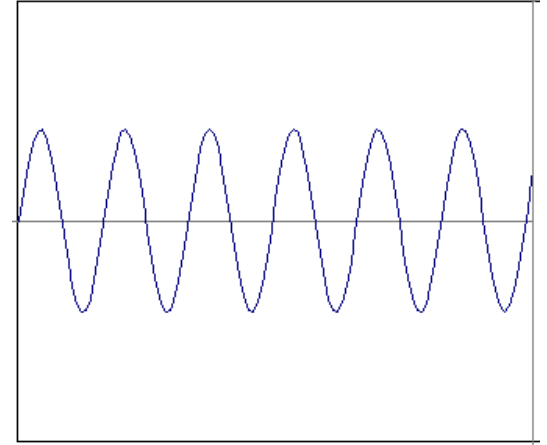
Frequency Spectra



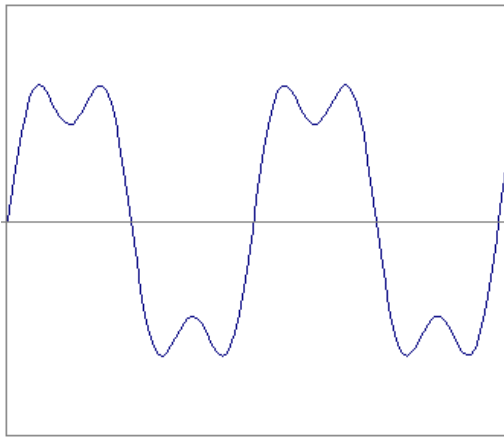
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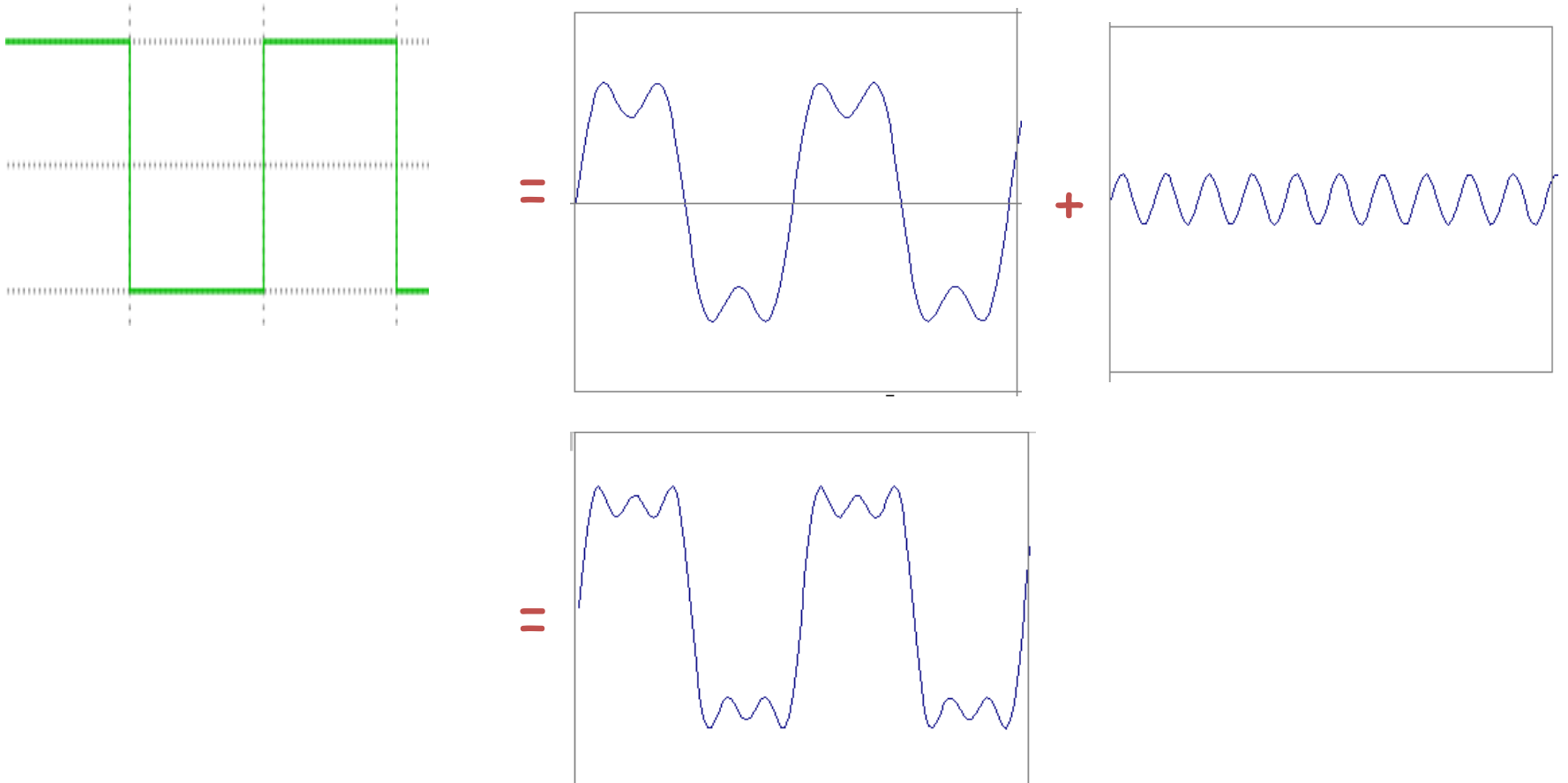
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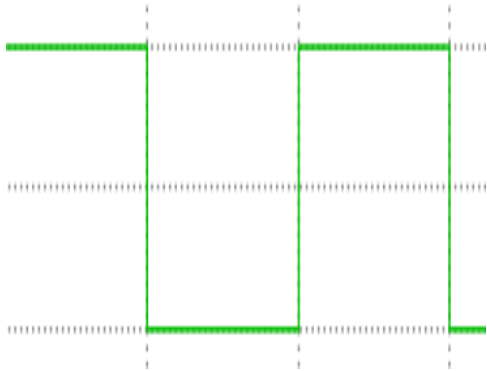
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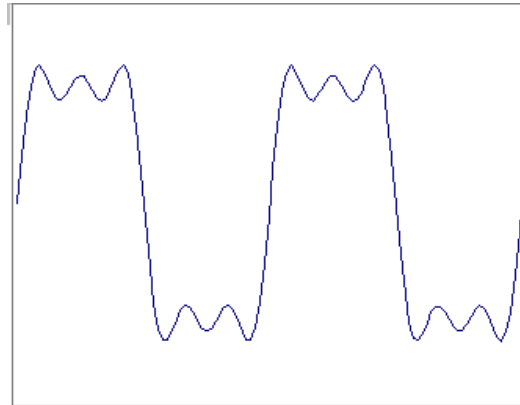
Frequency Spectra



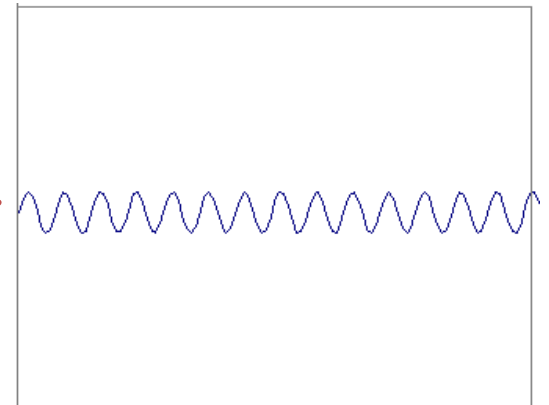
Frequency Spectra



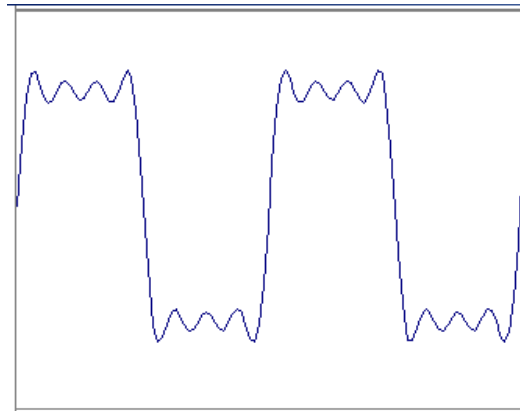
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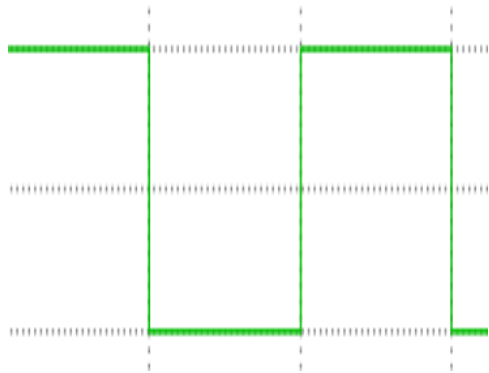
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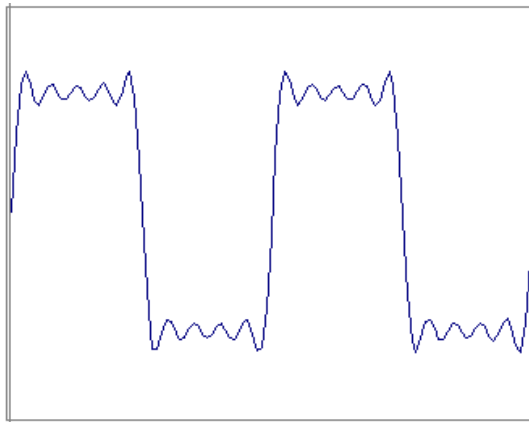
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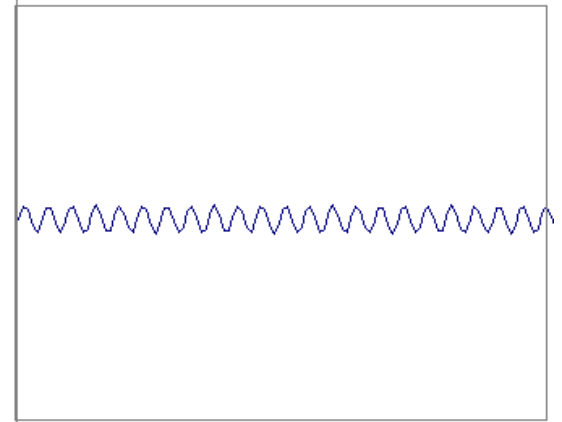
Frequency Spectra



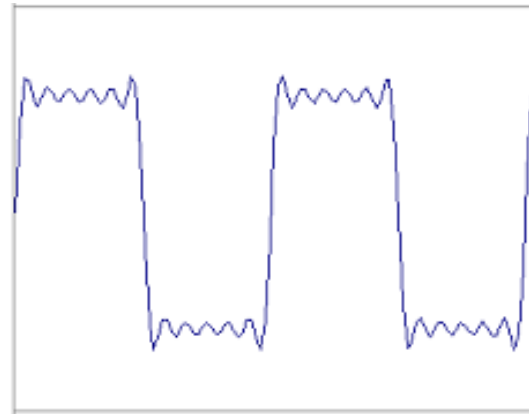
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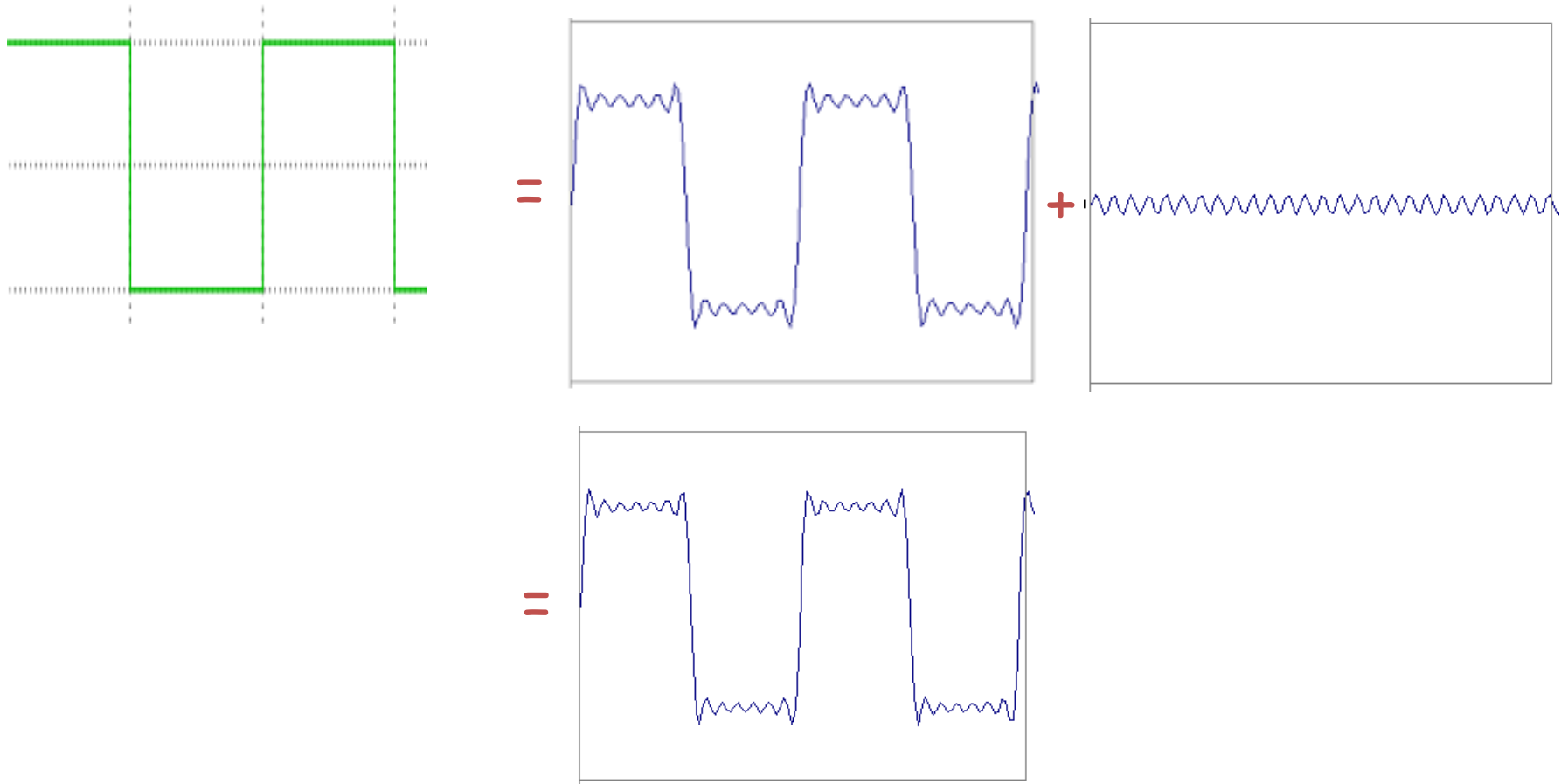
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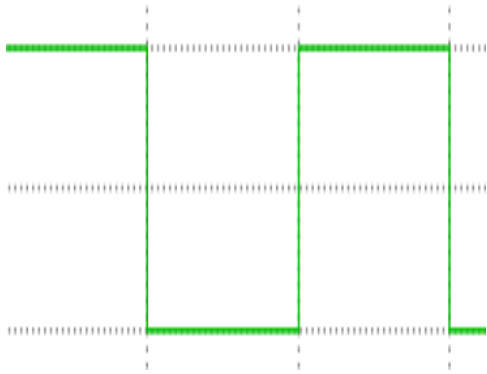
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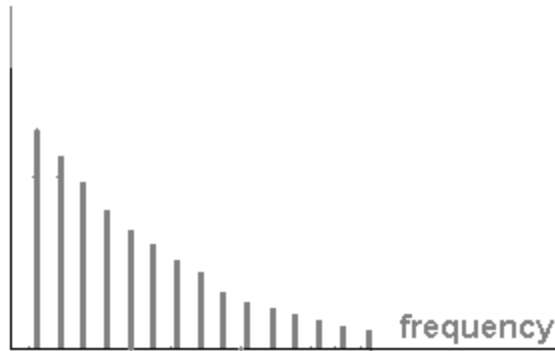
Frequency Spectra



Frequency Spectra

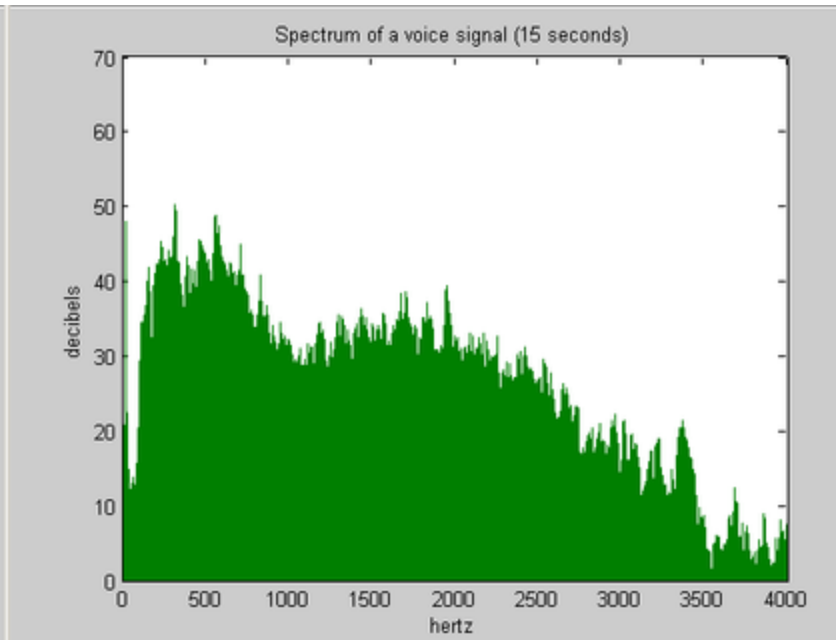
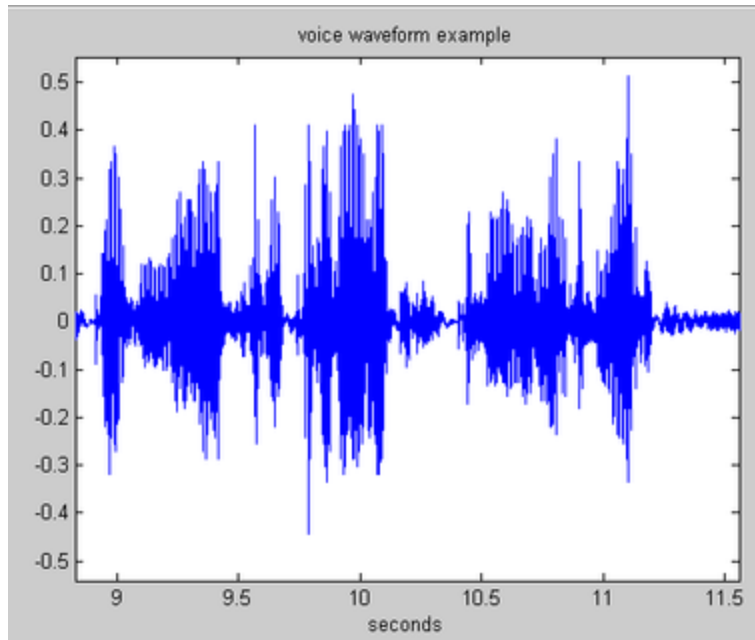


$$= A \sum_{k=1}^{\infty} \frac{1}{k} \sin(2\pi kt)$$



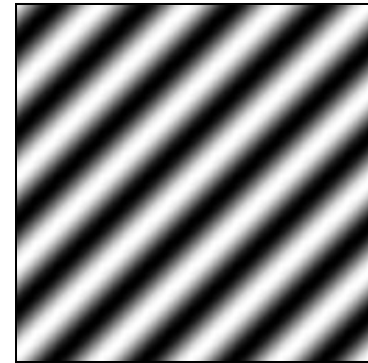
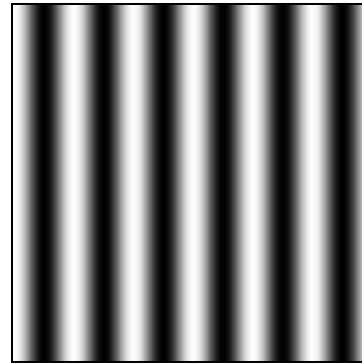
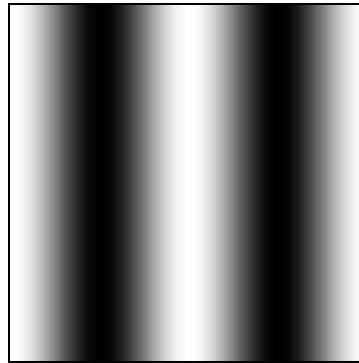
Example: Music

- We think of music in terms of frequencies at different magnitudes

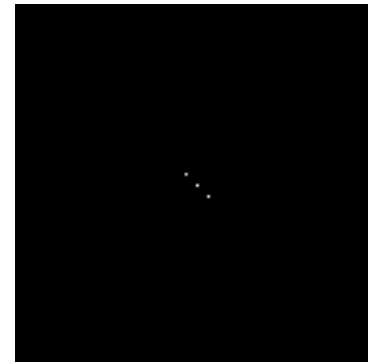
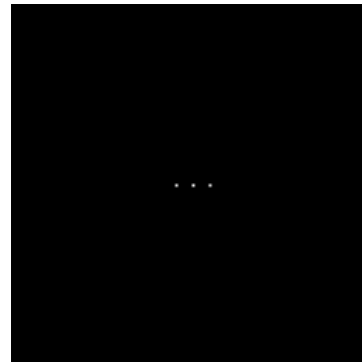
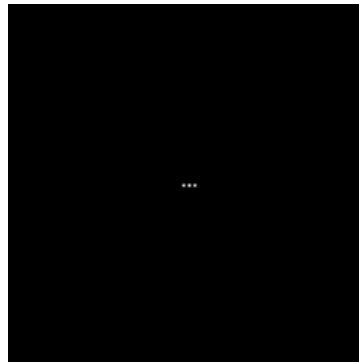


Fourier analysis in images

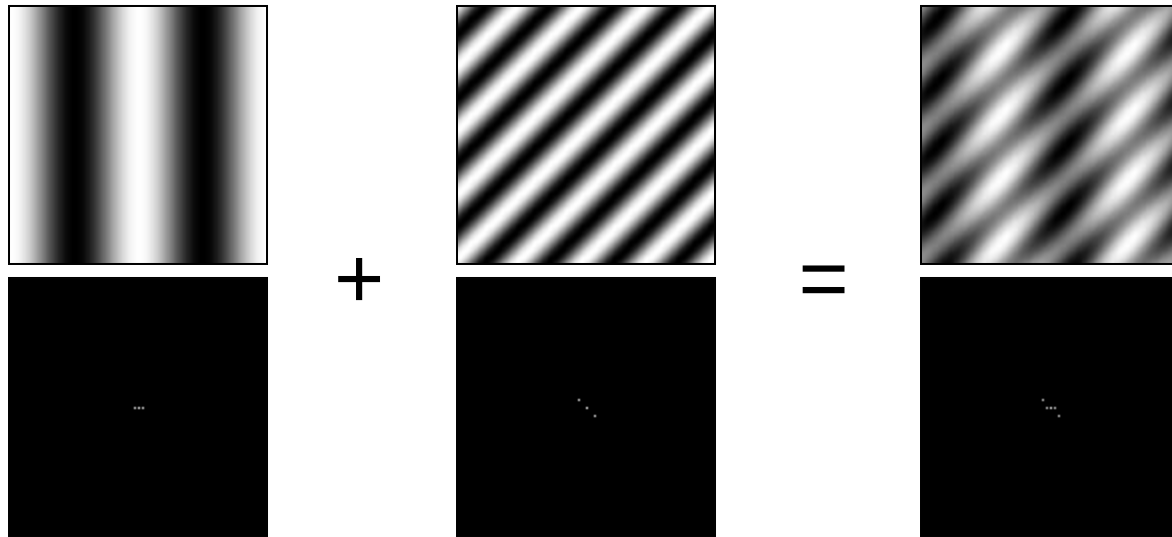
Intensity Image



Fourier Image



Signals can be composed



<http://sharp.bu.edu/~slehar/fourier/fourier.html#filtering>
More: <http://www.cs.unm.edu/~brayer/vision/fourier.html>

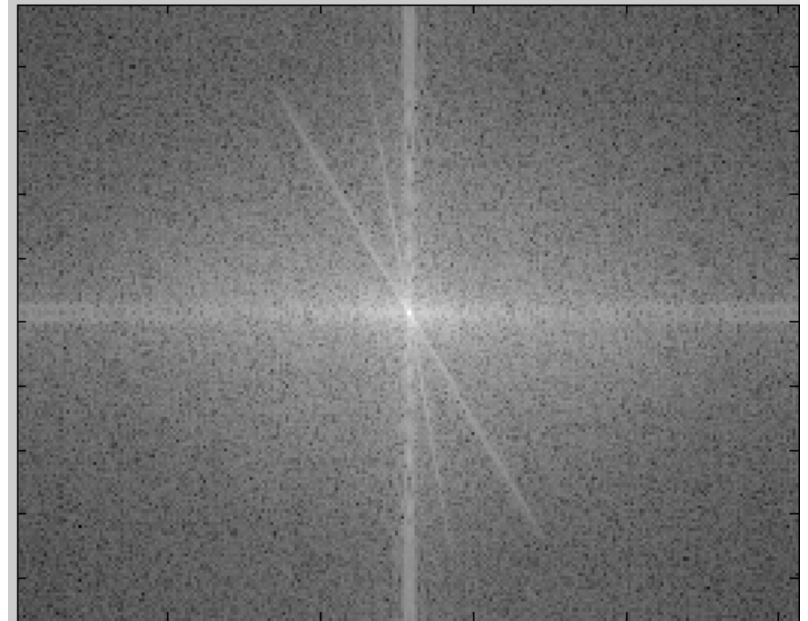
Fourier Transform

- Fourier transform stores the magnitude and phase at each frequency
 - Magnitude encodes how much signal there is at a particular frequency
 - Phase encodes spatial information (indirectly)
 - For mathematical convenience, this is often notated in terms of real and complex numbers

Amplitude: $A = \pm \sqrt{R(\omega)^2 + I(\omega)^2}$

Phase: $\phi = \tan^{-1} \frac{I(\omega)}{R(\omega)}$

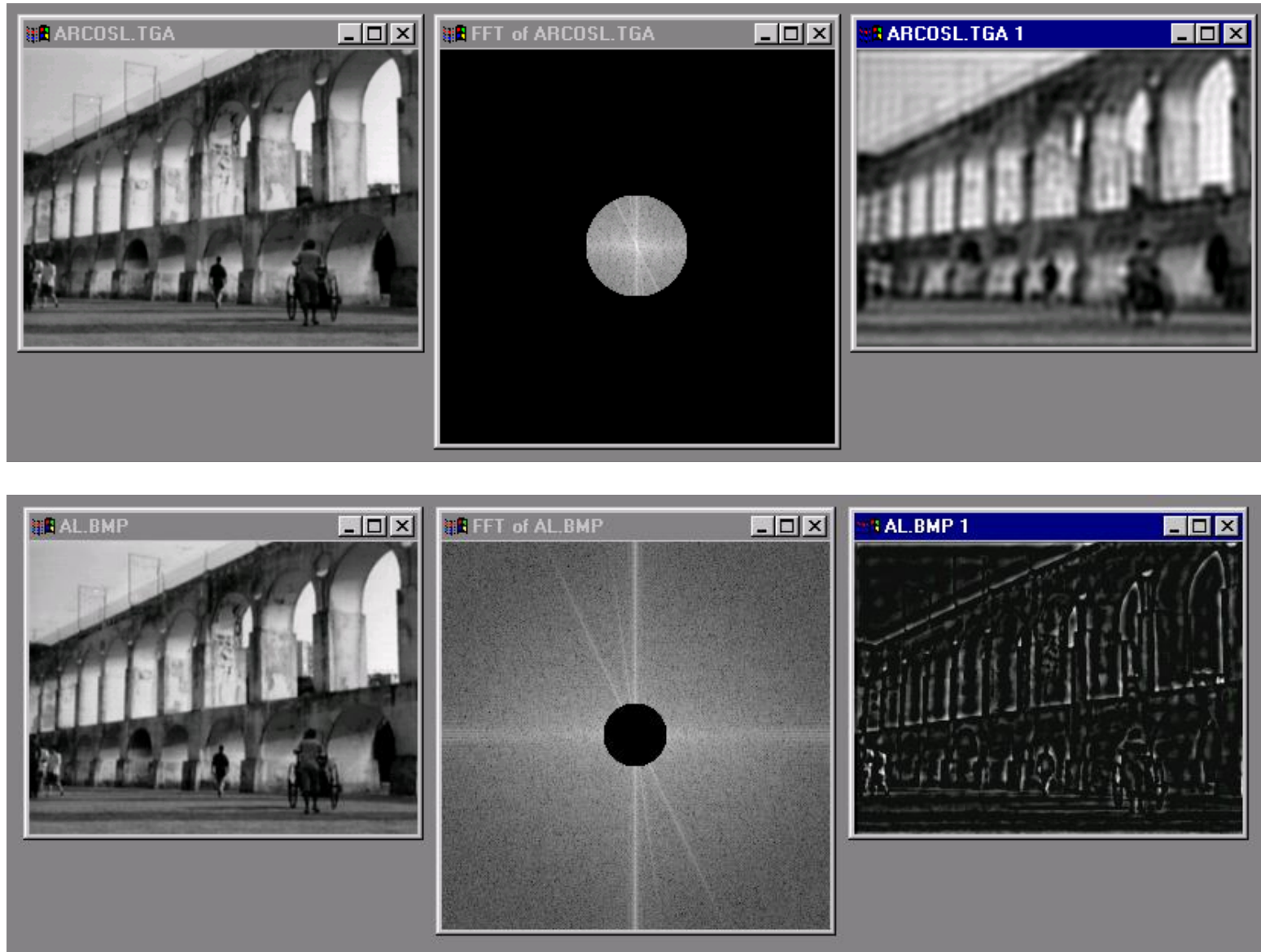
Man-made Scene



Can change spectrum, then reconstruct



Low and High Pass filtering



The Convolution Theorem

- The Fourier transform of the convolution of two functions is the product of their Fourier transforms

$$F[g * h] = F[g]F[h]$$

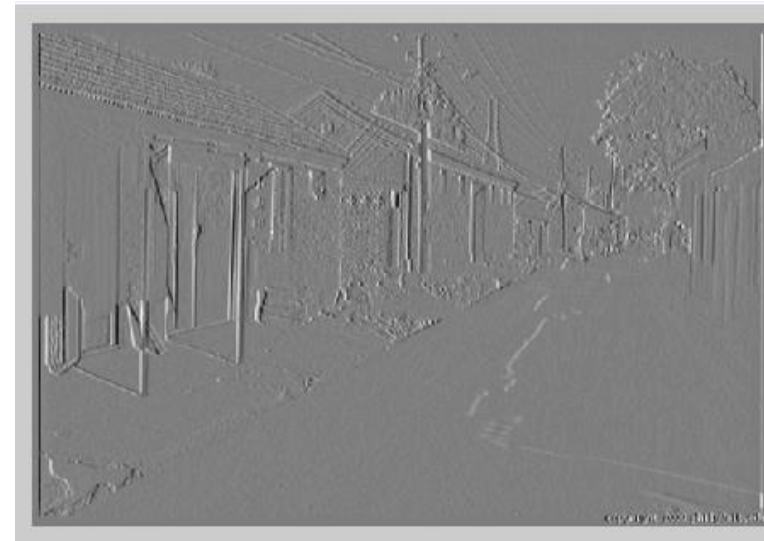
- **Convolution** in spatial domain is equivalent to **multiplication** in frequency domain!

$$g * h = F^{-1}[F[g]F[h]]$$

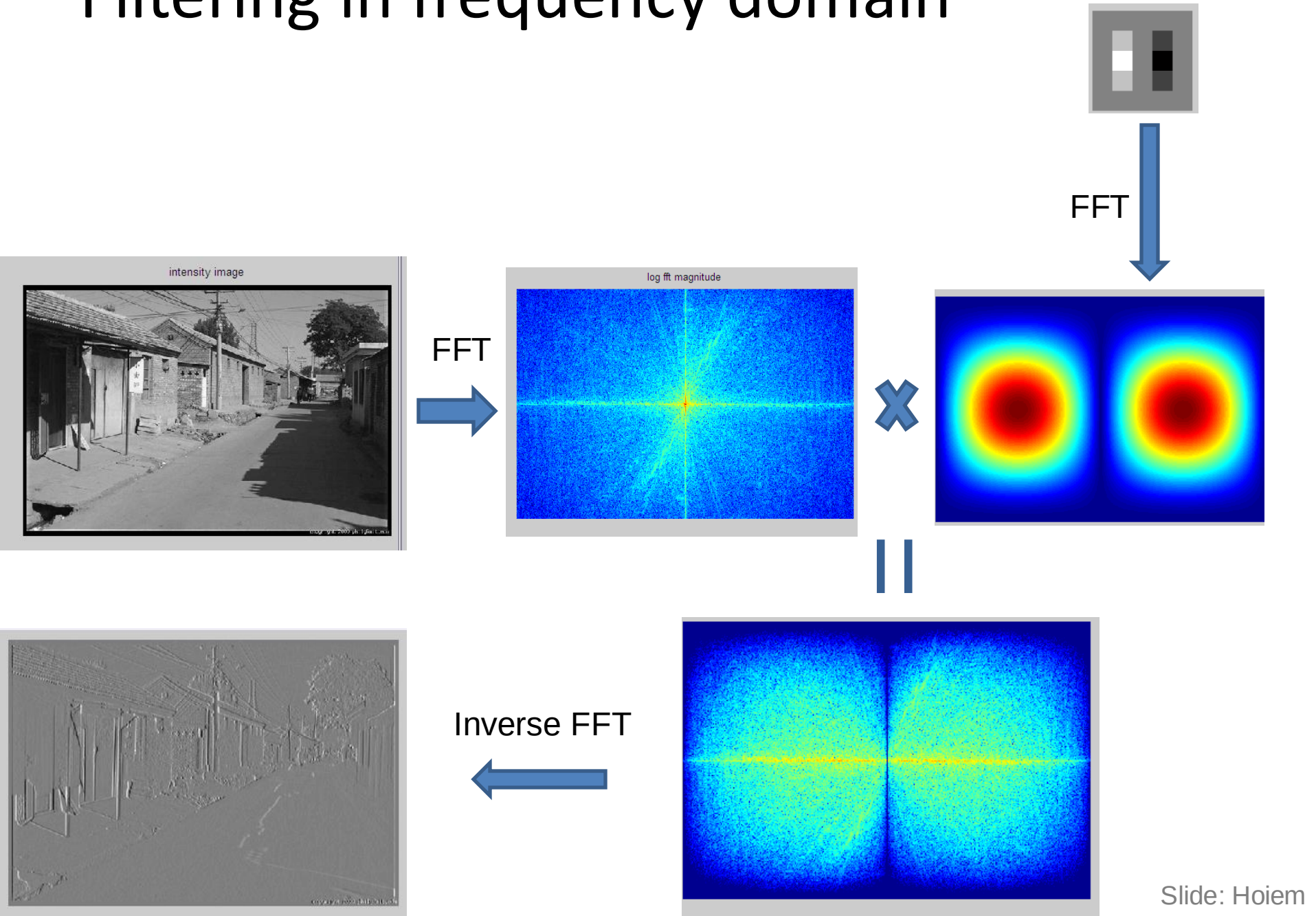
Filtering in spatial domain

1	0	-1
2	0	-2
1	0	-1

intensity image



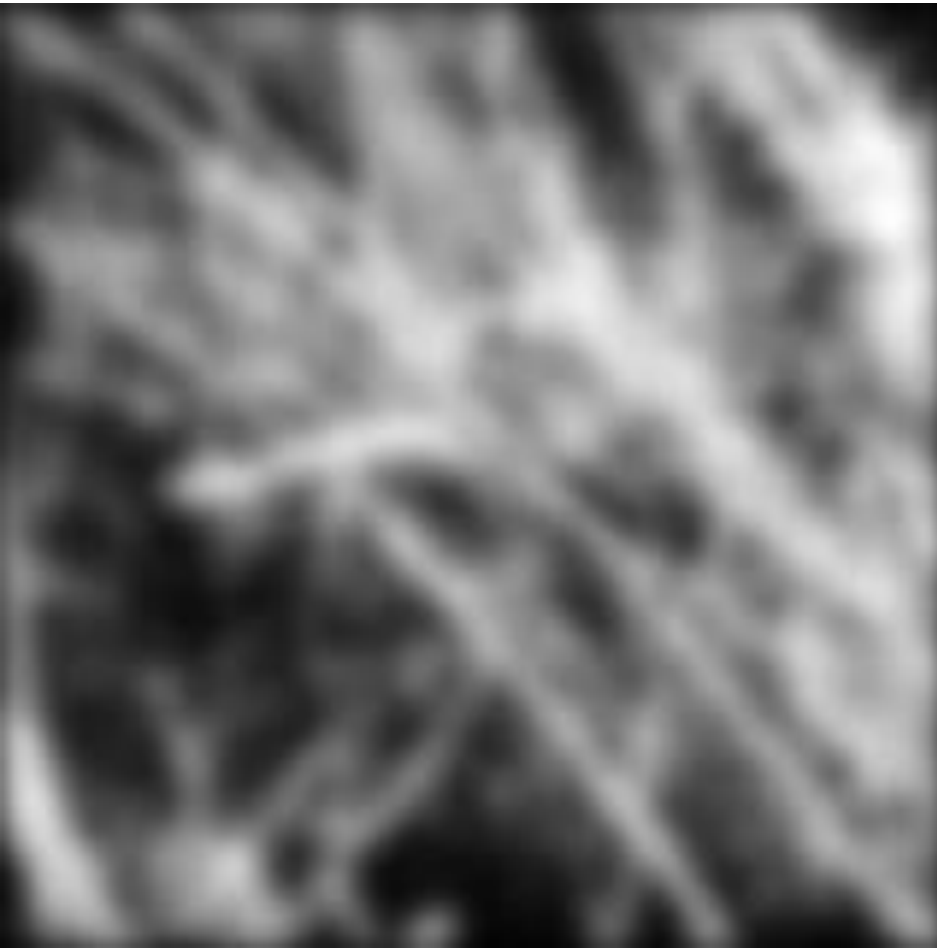
Filtering in frequency domain



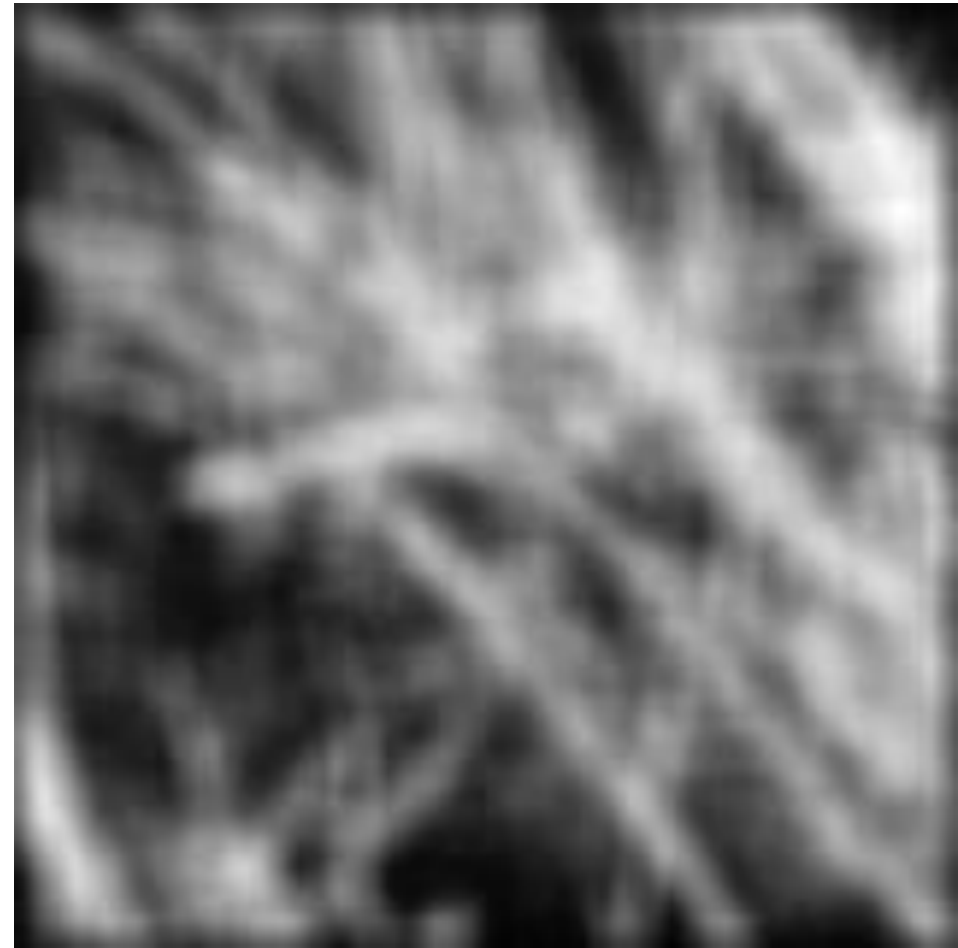
Filtering

Why does the Gaussian give a nice smooth image, but the square filter give edgy artifacts?

Gaussian



Box filter

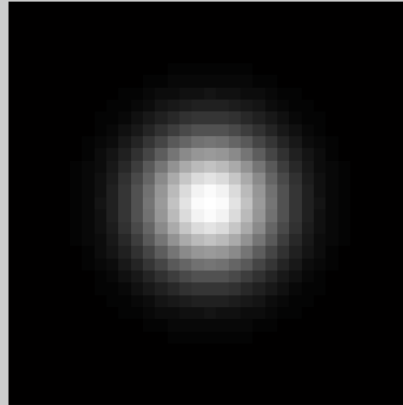


Gaussian

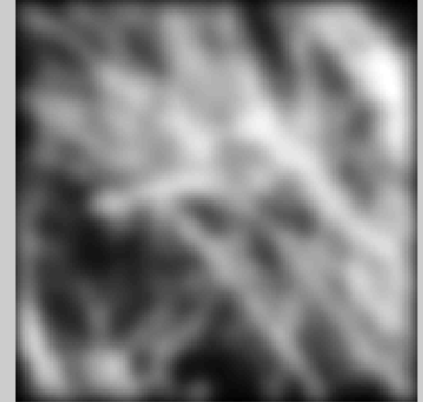
intensity image



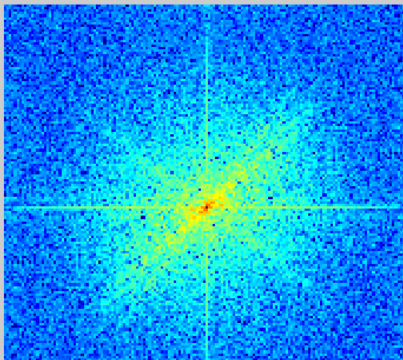
filter: gaussian



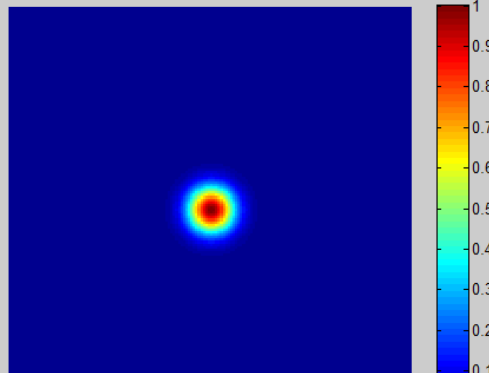
filtered image



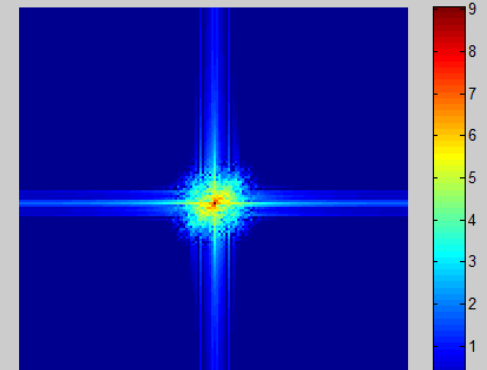
log fft magnitude of image



filter: gaussian



log fft magnitude of filtered image



Box Filter

